

Hippo Pathway in Organ Size Control, Tissue Homeostasis, and Cancer

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Two decades of studies in multiple model organisms have established the Hippo pathway as a key regulator of organ size and tissue homeostasis. By inhibiting YAP and TAZ transcription co-activators, the Hippo pathway regulates cell proliferation, apoptosis, and stemness in response to a wide range of extracellular and intracellular signals, including cell-cell contact, cell polarity, mechanical cues, ligands of G-protein-coupled receptors, and cellular energy status. Dysregulation of the Hippo pathway exerts a significant impact on cancer development. Further investigation of the functions and regulatory mechanisms of this pathway will help uncovering the mystery of organ size control and identify new targets for cancer treatment.

The emergence of multicellular organisms is an evolutionary milestone. Among the most fundamental mechanisms supporting multicellularity are those that ensure the proper size and shape of tissues and organs to meet the need of functionality. However, despite intensive investigations into the underlying principles behind a “preset” size of organs, we are far from having a clear picture of this basic question in developmental biology. Nevertheless, investigations of the Hippo pathway on organ size control have shed light into this mystery (Halder and Johnson, 2011; Pan, 2010; Yu and Guan, 2013).

In 1995, two studies in *Drosophila* discovered that deletion of *Warts* (*Wts*) gene resulted in dramatic overgrowth of multiple tissues (Justice et al., 1995; Xu et al., 1995). Several years later, a flurry of studies showed that *Salvador* (*sav*) (Kango-Singh et al., 2002; Tapon et al., 2002), *Hippo* (*hpo*) (Harvey et al., 2003; Jia et al., 2003; Pantalacci et al., 2003; Udan et al., 2003; Wu et al., 2003), and *Mob* as tumor suppressor (*mats*) (Lai et al., 2005) mutant *Drosophila* phenocopy *Wts* mutants with regards to tissue overgrowth. *Hpo*, *Sav*, *Wts*, and *Mats* interact genetically and physically, and the remarkable organ size phenotype elicited by their mutation is unprecedented in other established developmental pathways; thus, they were grouped into a new signaling module—the Hippo pathway—named after the enormous size of *hpo* mutant organs, which resembles that of a hippopotamus. *Yorkie* (*yki*), the key functional effector of the Hippo pathway in organ size regulation, was soon discovered in a screen for *Wts*-interacting proteins (Huang et al., 2005).

The Hippo pathway is highly conserved in mammals. The mammalian orthologs of *Hpo*, *Sav*, *Wts*, *Mats*, and *Yki* are *Mammalian sterile 20-like 1/2* (*MST1/2*, also called *STK4/3*), *Salvador* (*SAV1*), *Large tumor suppressor homolog 1/2* (*LATS1/2*), *MOB kinase activator 1A/B* (*MOB1a/b*), and *Yes-associated protein* (*YAP*)/transcriptional co-activator with PDZ binding motif (*TAZ*, also called *WWTR1*), respectively (Halder and Johnson, 2011; Pan, 2010; Yu and Guan, 2013) (Table 1). The Hippo

pathway quickly attracted broad attention due to its remarkable potency in regulating organ size, as well as its apparent relevance to tissue regeneration and cancer. Here, with an emphasis on recent developments, we review the current understanding of the organization, regulation, and function of the Hippo pathway and discuss some key open questions.

YAP/TAZ and Yki as Major Effectors of Hippo Signaling

In *Drosophila*, deleting *yki* suppresses the overgrowth phenotypes of *hpo*, *sav*, or *Wts* mutants (Huang et al., 2005). In mice, deleting *Yap* also diminishes the overgrowth phenotypes caused by deficiency of *Mst1/2* or other upstream regulators (Zhang et al., 2010; Zhou et al., 2011). Thus, *Yki* and *YAP/TAZ* are the evolutionarily conserved key effectors of the Hippo pathway.

Yki and *YAP/TAZ* are believed to mediate the biological functions of the Hippo pathway by regulating gene transcription. As transcriptional co-activators, *Yki* and *YAP/TAZ* cannot bind DNA directly, and they must interact with DNA-binding transcription factors to regulate target gene expression. In *Drosophila*, *Scalloped* (*Sd*) is a transcription factor and key partner of *Yki* that mediates the function of *Yki* in tissue growth (Goulev et al., 2008; Wu et al., 2008; Zhang et al., 2008; Zhao et al., 2008). Similarly, in mammalian cells, the TEAD family transcription factors (TEAD1–4, orthologs of *Sd*) are key partners for *YAP* (Zhao et al., 2008). Several lines of evidence indicate that TEADs are the major partners of *YAP* in transcriptional regulation. For instance, a TEAD-binding-deficient *YAP* mutant lost its ability to induce transcription of known *YAP* target genes (Zhao et al., 2008), and knockin of the TEAD-binding-deficient *YAP* (Y94A mutant) phenotypically mimics *YAP* knockout in the skin and heart (Schlegelmilch et al., 2011; von Gise et al., 2012). In addition, the majority of *YAP* and TEAD occupied sites in the genome are shared (Stein et al., 2015; Zanonato et al., 2015; Zhao et al., 2008), and when TEAD is fused with a VP16

Table 1. Hippo Pathway Components and Major Functions

Drosophila	Mammals	Major Functions in Hippo Pathway
Hippo (Hpo)	MST1/2	Phosphorylates LATS1/2, MOB1, and SAV1, leading to LATS1/2 activation
Salvador (Sav)	SAV1	Interacts with MST1/2, promotes phosphorylation of LATS1/2 by MST1/2
Warts (Wts)	LATS1/2	Phosphorylates and inactivates YAP/TAZ
Mats	MOB1A/B	Scaffold protein of LATS1/2
Yorkie (Yki)	YAP/TAZ	Transcription co-activator, major effector of the Hippo pathway
Scalloped (Sd)	TEAD1-4	Transcription factors that mediate the effect of YAP/TAZ
Tgi	VGLL4	Competes with YAP/TAZ for TEADs, inhibits YAP/TAZ functions
misshapen (Msn)	MAP4K4/6/7	Phosphorylates and activates LATS1/2
Merlin (Mer)	Merlin/NF2	May form a complex and mediates upstream signals (from plasma membrane) to MST1/2; NF2 may bring LATS1/2 to plasma membrane and facilitate its activation by MST1/2
Kibra	KIBRA	
Expanded (Ex)	FRMD6?	Sequesters YAP/TAZ to cell junctions, binding and indirectly activating LATS1/2; a substrate of LATS1/2
	AMOT	

transactivation domain, it produces a gene expression profile similar to that induced by active YAP (Ota and Sasaki, 2008). TEAD1–4 or Sd can bind to a consensus motif similar to the GTIIC sequence (TGGAATGT or ACATTCCA), and transcription reporters under control of GTIIC concatemers are now widely used to measure Hippo pathway activity (Dupont et al., 2011; Mohseni et al., 2014; Ota and Sasaki, 2008). Besides TEAD1–4, YAP/TAZ have also been shown to interact with other transcription factors, including Smad, RUNX1/2, p63/p73, and ErbB4 (reviewed in Varelas, 2014), although the functional significance of these transcription factors in Hippo pathway is less clear.

A strong transcriptional activation domain is present in YAP/TAZ but absent in Yki. Nevertheless, human YAP can rescue the lethality resulting from Hippo pathway hyperactivation in *Drosophila*, indicating a functional conservation (Huang et al., 2005). Genome-wide assessment of chromatin binding status reveals that, in addition to occupancy at proximal promoters of target genes, YAP and TEAD largely exert their transcriptional activity by interacting with distal enhancers, suggesting that YAP, and probably TAZ and Yki, may regulate transcription via multiple mechanisms, including recruitment of general transcription factors, modification of epigenetic markers, and modulation of chromatin structure (Lian et al., 2010; Stein et al., 2015; Zancanato et al., 2015) (Figure 1). Indeed, recent evidence shows that Yki can interact with the Brahma complex, GAGA factors, nuclear receptor coactivator 6 (NCoA6, a subunit of the Trithorax-related histone methyltransferase), and the Mediator complex (Jin et al., 2013; Oh et al., 2013; Qing et al., 2014), and TAZ can interact with the chromatin-remodeling complex SWI/SNF (Skibinski et al., 2014).

The transcriptional activity of Yki and YAP/TAZ is regulated in the nucleus by Tondue-domain-containing growth inhibitor (Tgi) and Vestigial-like family member 4 (VGLL4, an ortholog of Tgi). Tgi can directly compete with Yki for Sd binding, resulting in inhibition of Yki-regulated transcription (Guo et al., 2013; Koontz et al., 2013). When Hippo signaling is on, Tgi and Sd form a complex, leading to transcriptional repression; on the contrary, when Hippo signaling is off, Yki enters the nucleus and displaces Tgi

from Sd, leading to expression of Yki target genes and tissue growth (Koontz et al., 2013). In mammals, VGLL4 similarly competes with YAP/TAZ for TEAD binding (Jiao et al., 2014; Zhang et al., 2014b) (Figure 1). However, whether YAP/TAZ functions simply by relieving a default repression of TEAD by VGLL4 has not been demonstrated. Interestingly, the expression of VGLL4 appears to be repressed by miR-130a—a microRNA (miRNA) that is directly induced by YAP, leading to amplification of YAP activity (Shen et al., 2015). A similar mechanism is also present in *Drosophila*, in which *Bantam*—a well-known Yki-induced miRNA (Nolo et al., 2006; Thompson and Cohen, 2006), can repress expression of Tgi (Shen et al., 2015).

Although their role as transcriptional co-activators is widely appreciated, YAP/TAZ and Yki may also repress expression of certain genes when bound to specific factors. For instance, YAP/TAZ can interact with the nucleosome-remodeling and histone deacetylase (NuRD) complex, resulting in transcriptional repression (Kim et al., 2015). Moreover, YAP has been identified as a regulator for global miRNA biogenesis via modulation of miRNA-processing enzymes Microprocessor or Dicer complex, suggesting a transcription-independent role for YAP (Chaulk et al., 2014; Mori et al., 2014). Hence, YAP/TAZ and Yki may regulate gene expression via multiple mechanisms.

Gene expression signatures under YAP/TAZ and Yki hyperactivation (or ectopic expression) in *Drosophila* and different mammalian cell types have been profiled by independent studies. However, the overlap between these different gene profiling studies is not high, suggesting that YAP/TAZ and Yki may regulate target gene expression in a tissue- or cell-type-specific manner. In *Drosophila*, some common Yki target genes are *Bantam*, *diap1*, and *cyclin E*, which may mediate the effect of Yki in inhibiting cell death and promoting cell proliferation (Huang et al., 2005; Nolo et al., 2006; Tapon et al., 2002; Thompson and Cohen, 2006). In mammals, connective tissue growth factor (CTGF) is a commonly used marker of YAP activation (Zhao et al., 2008), and many genes involved in cell proliferation, cell adhesion, and cell migration have also been identified as YAP targets (Stein et al., 2015). It is likely that a panel of genes work together to exert biological functions of YAP/TAZ.

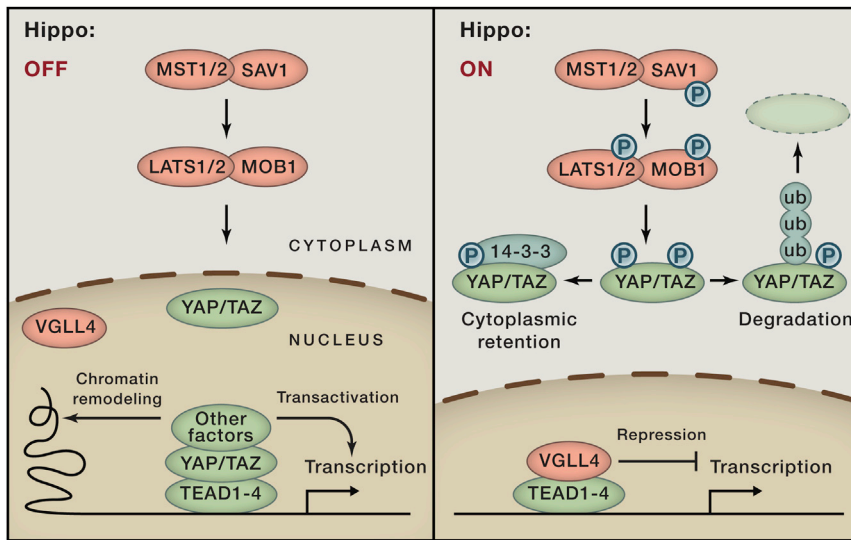


Figure 1. Inhibition of YAP/TAZ Transcriptional Coactivators by LATS1/2

(Left) When Hippo signaling is off, YAP/TAZ enter the nucleus, compete with VGLL4 for TEADs, and recruit other factors to induce gene transcription. YAP/TAZ may bind proximal promoters or distal enhancers of target genes to induce transcription. (Right) When Hippo signaling is on, YAP/TAZ are phosphorylated by LATS1/2 on multiple sites, resulting in interaction with 14-3-3 and cytoplasmic retention; phosphorylation also leads to YAP/TAZ poly-ubiquitination and degradation. VGLL4 interacts with TEADs and represses target gene transcription.

et al., 2002) (Figure 1). In *Drosophila*, Wts is regulated by Hpo, Sav, and Mats by a similar mechanism (Wehr et al., 2013).

Merlin (Mer), Expanded (Ex), and Kibra, three cell-cortex-localized and cytoskeleton-interacting proteins, may function

as a scaffold for core Hippo components at the apical domain for activation, as Sav, Hpo, and Wts have been shown to physically interact with Ex/Mer/Kibra (Baumgartner et al., 2010; Genet et al., 2010; Hamaratoglu et al., 2006; Yu et al., 2010) (Table 1). In addition, the effect of Ex/Mer/Kibra on the Hippo pathway may be mediated by Tao kinase 1 (Tao-1), which can directly phosphorylate and activate Hpo (Boggiano et al., 2011; Poon et al., 2011). In mammalian cells, Neurofibromin 2 (NF2, Mer ortholog) appears to play a more direct role in regulating LATS1/2 activity: it can directly interact with LATS1/2 and recruit LATS1/2 to the plasma membrane for activation by MST1 (Yin et al., 2013; Zhang et al., 2010) (Figures 2 and 3).

Hpo or MST1/2 are not absolutely required for regulation of Wts or LATS1/2. It has been observed that, in mouse embryonic fibroblast (MEF) cells, MST1/2 double knockout did not abolish YAP phosphorylation, suggesting the existence of additional Hippo-like activity (Zhou et al., 2009). Indeed, a recent study in *Drosophila* has identified Misshapen (Msn) as another kinase responsible for Wts activation. This mechanism is also conserved in mammals, as MAP4K4 (Msn ortholog) overexpression promotes phosphorylation of LATS1/2 (Li et al., 2014), and MAP4K4 knockdown induces activity of a YAP reporter (Mohseni et al., 2014). In addition to MAP4K4, two recent studies revealed that many MAP4K family kinases, including MAP4K1/2/3/5 (Happyhour in *Drosophila*) and MAP4K4/6/7 (Msn in *Drosophila*), can directly phosphorylate and activate LATS1/2 (Meng et al., 2015; Zheng et al., 2015). These kinases, together with MST1/2, may regulate LATS1/2 activity in a tissue- and signal-specific manner. It is also possible that additional kinases, especially some STE20 family members, may activate LATS1/2 in response to different upstream signals or in different tissue contexts (Figures 2 and 3).

YAP/TAZ have also been shown to be phosphorylated by many other kinases such as cyclin-dependent kinase 1 (CDK1), Jun N-terminal kinases (JNK), homeodomain-interacting protein kinases (HIPK), ABL, and Src family tyrosine kinases (reviewed in Varelas, 2014), suggesting that YAP/TAZ can be

MST1/2 and LATS1/2 Kinases Restrict YAP Activity

In the Hippo pathway, LATS1/2 directly phosphorylate and inhibit YAP/TAZ. Interestingly, YAP has five LATS1/2 target consensus motifs (HXRXXS), four of which are conserved in TAZ (Zhao et al., 2010). Phosphorylation of YAP on serine 127 (S127) generates a 14-3-3 binding site, and binding with 14-3-3 sequesters YAP in the cytoplasm (Dong et al., 2007; Zhao et al., 2007). In addition, phosphorylation of YAP on serine 381 (S381) triggers a subsequent phosphorylation by casein kinase 1 (CK1 δ/ϵ) and activation of a phosphodegron, resulting in recruitment of SCF ^{β -TRCP} E3 ligase, ubiquitination, and proteasomal degradation of YAP (Zhao et al., 2010). Thus, through regulating both YAP subcellular localization and protein stability, LATS1/2 ensures a spatial and temporal control of YAP activity (Figure 1). TAZ is regulated by LATS1/2 in a similar fashion, although degradation plays a more prominent role in TAZ regulation possibly due to an additional phosphodegron at its N terminus (Lei et al., 2008; Liu et al., 2010a). The subcellular localization of Yki is regulated similarly by Wts phosphorylation and 14-3-3 binding. However, the phosphodegron and the phosphorylation-mediated degradation mechanisms are not conserved in Yki (Huang et al., 2005; Oh and Irvine, 2008). In addition, YAP and Yki have also been shown to be degraded via the autolysosomal pathway (Kwon et al., 2013; Liang et al., 2014), suggesting a potential role of YAP and Yki in vesicular membrane dynamics and related cellular processes such as autophagy.

LATS1/2 are activated by MST1/2 through several mechanisms. MST1/2 phosphorylate LATS1/2 at the C-terminal hydrophobic motif, which promotes LATS1/2 autophosphorylation at its activation loop. Furthermore, phosphorylation of MOB1 by MST1/2 enhances MOB1 interaction with the autoinhibitory domain of LATS1/2, leading to full activation of LATS1/2 (Callus et al., 2006; Chan et al., 2005; Praskova et al., 2004; Wu et al., 2003). In addition, SAV1 is also phosphorylated by MST1/2, and SAV1 functions as a partner of MST1/2 in promoting LATS1/2 phosphorylation (Callus et al., 2006; Tapon

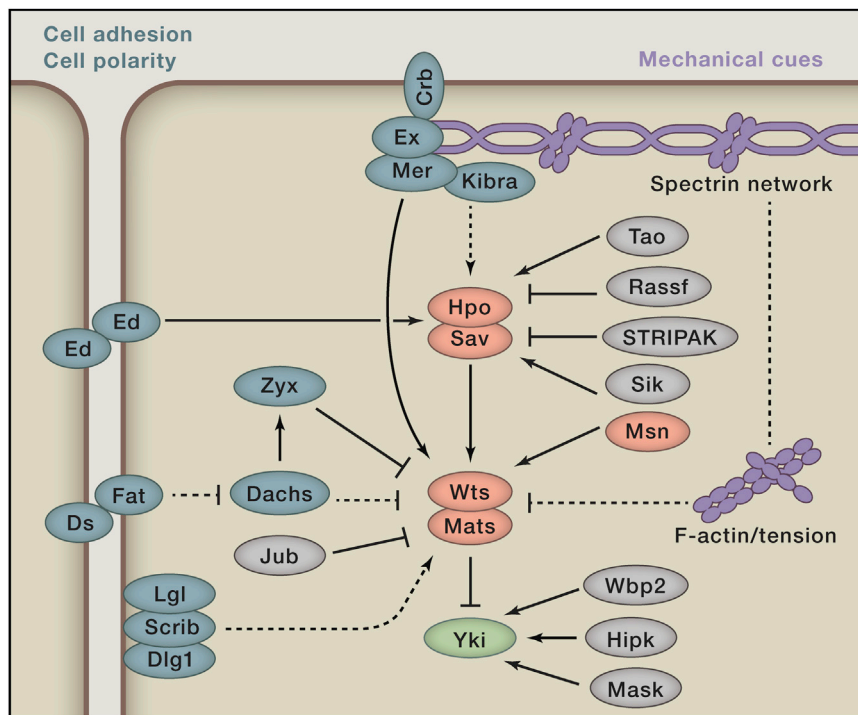


Figure 2. Regulation of the Hippo Pathway in *Drosophila*

The Hippo pathway is regulated by cell-adhesion molecules (Ed), determinants of cell polarity (Crb, Fat/Ds, Scrib complex), and mechanical cues (spectrin, F-actin, or cellular tension). In addition, Hpo is regulated by Tao, salt-induced kinase (Sik), Ras-associated factor (Rassf), striatin-interacting phosphatase and kinase (STRIPAK) complex; Wts is regulated by Zyxin (Zyx) and Jub; Yki is regulated by WW-domain-binding protein 2 (Wbp2), Hipk, and multiple ankyrin repeats single-KH domain (Mask) (refer to Varelas, 2014). Arrows, blunt ends, and dashed lines indicate activation, inhibition, and indirect regulation, respectively.

Cell Contact and Mechanical Cues Regulate Hippo Signaling

Cells in solid tissues communicate with neighboring cells and their extracellular matrix (ECM) and perceive constant physical signals from their local environment. Cell-cell contact was discovered as the first signal regulating the Hippo pathway (Zhao et al., 2007). In a sparse culture, YAP/TAZ are primarily localized in the nucleus to promote target gene transcription and cell proliferation. On

regulated by mechanisms independent of Hippo pathway kinases.

Cell Polarity and Cell Adhesion Regulate Hippo Signaling

In searching for upstream regulators of the Hippo pathway, many proteins involved in cell polarity and cell adhesion have been identified. Echinoid (Ed), a cell adhesion molecule in *Drosophila*, can interact with and stabilize Sav and leads to activation of Hpo (Yue et al., 2012). In mammalian cells, several proteins at adherens and tight junctions, such as Angiomotin (AMOT), protein tyrosine phosphatase non-receptor type 14 (PTPN14), and α -catenin, can sequester YAP/TAZ at cell junctions (reviewed in Yu and Guan, 2013). Therefore, cell adhesion and formation of intercellular junctions serve as a mechanism to repress YAP/TAZ transcriptional activity (Figures 2 and 3).

Crumbs (Crb), a component of apical-basal polarity, interacts with Ex, which is critical for the apical localization of Ex/Mer/Kibra complex (Chen et al., 2010; Ling et al., 2010; Robinson et al., 2010). In addition, Scribble (SCRIB) interacts with both MST1/2 and LATS1/2, thus promoting LATS1/2 activation (Cordenosi et al., 2011; Mohseni et al., 2014). Fat, a protocadherin that plays a key role in planar cell polarity, also activates the Hippo pathway, possibly through regulating Ex protein levels and localization (Bennett and Harvey, 2006; Silva et al., 2006; Tyler and Baker, 2007; Willecke et al., 2006) (Figure 2). However, mammalian Fat orthologs do not seem to be major regulators of the Hippo pathway (Sharma and McNeill, 2013). It is noteworthy that the link between cell polarity and the Hippo pathway may be indirect, and some proteins, such as Fat, may regulate cell polarity and Hippo signaling via different mechanisms (Matakatsu and Blair, 2012).

the contrary, at high cell density, YAP/TAZ are primarily cytoplasmic, corresponding to growth inhibition. It is known that a cell ceases to proliferate following physical contact with surrounding cells, and loss of cell contact inhibition is an indicator of oncogenic transformation. Thus, regulation of YAP/TAZ by cell density suggests a critical role for the Hippo pathway in contact inhibition, tissue growth, and tumorigenesis.

Mechanical cues, such as ECM stiffness and cell geometry, are also potent regulators of YAP/TAZ (Dupont et al., 2011) (Figure 3). When cells are grown on a stiff matrix or are spread across a large surface, YAP/TAZ are activated. In contrast, when cells are seeded on a soft matrix or are compressed into a small area, YAP/TAZ are inactivated. ECM stiffness and cell geometry are important for cell proliferation and differentiation, and YAP/TAZ activity plays a role in these cellular processes. Cell geometry has also been proposed as a mechanism underlying Hippo pathway regulation by cell density: at low density, cells are flat and spread, leading to YAP activation; whereas at high density, cells adopt a round and compact geometry, resulting in YAP inactivation (Aragona et al., 2013; Wada et al., 2011). As further support for a role of YAP/TAZ in mechanosensing, mechanical strain and shear stress have been shown to stimulate YAP/TAZ, and YAP activation is required for mechanical strain-induced cell-cycle reentry (Benham-Pyle et al., 2015; Codellia et al., 2014; Kim et al., 2014). In a medaka fish model, YAP also mediates changes in three-dimensional body shape in response to tissue tension (Porazinski et al., 2015).

The attachment of cells to the ECM is an important mechanism for maintaining cell survival, and cells normally undergo anoikis when detached from the ECM. The attachment status of a cell to its ECM can also regulate Hippo pathway activity. YAP is

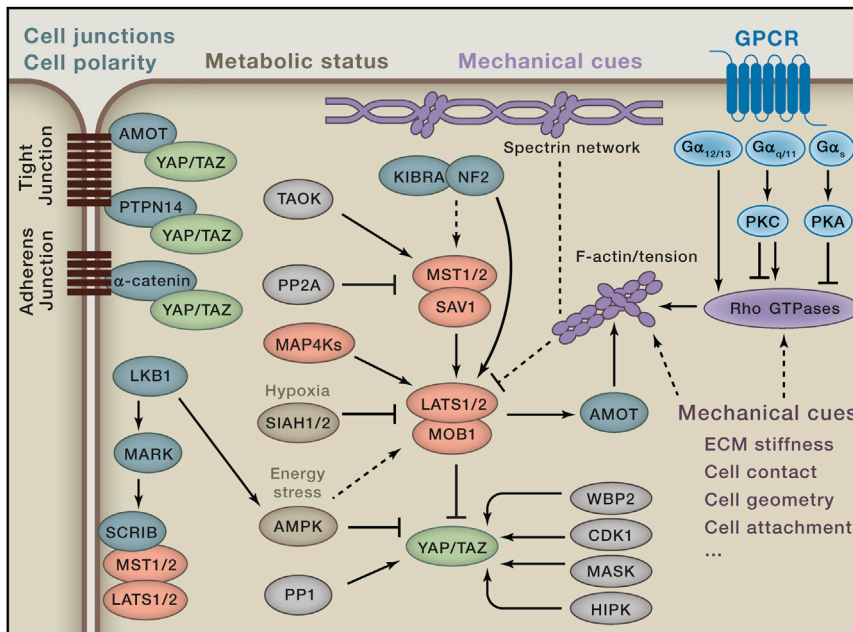


Figure 3. Regulation of the Hippo Pathway in Mammals

The Hippo pathway is regulated by diverse signals: (1) determinants of cell polarity and cell-cell junctions, such as SCRIB, which interacts with MST1/2 and LATS1/2, AMOT, PTPN14, and α -catenin, which can sequester YAP/TAZ to cell junctions; (2) mechanical cues, such as stiffness, cell contact, cell geometry, and cell attachment status that regulate the Hippo pathway by modulating activity of Rho GTPases, remodeling the actin cytoskeleton, or altering cellular tension; both apical and basolateral spectrin networks may function as sensors for mechanical cues in Hippo pathway regulation; (3) soluble factors, especially ligands for GPCRs, regulate LATS1/2 likely through Rho GTPases and actin dynamics; (4) metabolic status, such as cellular energy and oxygen stress, also regulate Hippo signaling; many other proteins such as protein phosphatase 2A (PP2A), protein phosphatase 1 (PP1), WBP2, CDK1, MASK, and HIPK can also regulate activities of different Hippo pathway components (refer to Varelas, 2014). Arrows, blunt ends, and dashed lines indicate activation, inhibition, and indirect regulation, respectively.

activated during the process of cell attachment but inactivated when cells are detached (Zhao et al., 2012). Expression of constitutively active YAP promotes survival of detached cells, suggesting that cancer cells with high YAP activity may escape anoikis and undergo metastasis. In support of this notion, LATS1/2 expression was found to be selectively reduced in metastatic, but not primary, prostate tumors (Zhao et al., 2012).

Soluble Factors Regulate Hippo Signaling

The primary function of YAP/TAZ is to promote growth, and many mitogenic hormones and growth factors act through G-protein-coupled receptors (GPCR) to induce cell proliferation (Figure 3). Ligands signal through GPCRs coupled to $G_{\alpha_{12/13}}$ or $G_{\alpha_{q/11}}$, such as lysophosphatidic acid, thrombin, angiotensin II, and estrogen, and can activate YAP/TAZ; in contrast, ligands signal through G_{α_s} -coupled GPCRs and protein kinase A (PKA), such as epinephrine and glucagon, can repress YAP/TAZ activity (Kim et al., 2013; Miller et al., 2012; Mo et al., 2012; Wennmann et al., 2014; Yu et al., 2013a; Yu et al., 2012; Zhou et al., 2015). Interestingly, activation of protein kinase C (PKC) by $G_{\alpha_{q/11}}$ can either activate or inhibit YAP, with conventional PKC activating YAP and novel PKC inhibiting YAP (Gong et al., 2015). The remarkable differential functions of PKC in YAP regulation provide a mechanism to explain some of the cell-type-specific responses to PKC activation. GPCR is the largest family of membrane receptors mediating diverse physiological or pathological responses. The demonstration of Hippo regulation by GPCRs links the Hippo pathway to a wide range of upstream signals and biological functions.

The Wnt/ β -catenin pathway is a key signaling cascade in development and carcinogenesis. A destruction complex including Axin, adenomatous polyposis coli (APC), and glycogen synthase kinase-3 (GSK3) causes constant degradation of β -catenin. Wnt stimulation disrupts the destruction complex and leads to accumulation of β -catenin. Interestingly, recent studies sug-

gest that YAP/TAZ are also activated by diverse Wnt family ligands. YAP/TAZ have been shown to be components of the destruction complex and are regulated by Wnt in a fashion similar to that of β -catenin (Azzolin et al., 2012, 2014). However, a recent study suggests that Wnt activates YAP/TAZ via Frizzled (a GPCR-like Wnt receptor), $G_{\alpha_{12/13}}$, Rho GTPases, and LATS1/2 (Park et al., 2015). In addition, APC has been shown to act as a scaffold protein for SAV1 and LATS1, and Apc deletion leads to YAP activation and tumorigenesis (Cai et al., 2015). More studies may be needed to verify the mechanism of YAP/TAZ activation by Wnt.

Epidermal growth factor (EGF) and insulin have also been shown to regulate YAP/Yki activity in cultured mammalian cells and *Drosophila*, and these signals are mediated by the Ras-Raf-MAPK signaling cascade or phosphoinositide-dependent kinase (PDK1) (Fan et al., 2013; Reddy and Irvine, 2013; Straßburger et al., 2012). TAZ is also stabilized upon PI3K activation, which is mediated by direct phosphorylation by GSK3 (Huang et al., 2012). However, no significant effect of EGF and IGF on YAP was observed in several other studies, and YAP activity appeared to be normal in the presence of inhibitors of PI3K or AKT or in PDK1 null embryonic cells (Yu et al., 2012; Zhao et al., 2007). These discrepancies could be due to differences in experimental settings or cell types and should be clarified by future studies.

Effect of Cellular Metabolic Status on Hippo Signaling

Recently, a link between cellular metabolic status and the Hippo pathway has been reported (Figure 3). Under energy deprivation, such as glucose starvation, the AMPK-activated protein kinase (AMPK) can directly phosphorylate YAP at serine 61 (S61) and serine 94 (S94) (Mo et al., 2015; Wang et al., 2015). Phosphorylation of YAP S94 abolishes the interaction between YAP and TEADs, leading to inhibition of YAP activity. In addition, energy stress also inhibits YAP by increasing kinase activity of LATS1/2 either in an AMPK-dependent or -independent

manner (Mo et al., 2015; Wang et al., 2015). AMPK can phosphorylate AMOTL1, which in turn facilitates YAP phosphorylation by LATS1/2 (DeRan et al., 2014). A similar mechanism is also present in *Drosophila*, in which Ampk inactivates Yki and affects cell proliferation in the larval central brain and central nerve cord (Gailite et al., 2015). It appears that, during energy stress, both AMPK and LATS1/2 are unleashed to restrict YAP activity. These findings also suggest that the Hippo pathway may mediate the anticancer effect of metformin, which is known to activate AMPK (DeRan et al., 2014; Mo et al., 2015). Other than AMPK, glucose may also promote YAP/TAZ activity through phosphofructokinase, which stimulates the interaction between YAP/TAZ and TEADs (Enzo et al., 2015).

YAP/TAZ activity has also been linked to oxygen availability. Under hypoxic conditions, the hypoxia-inducible factor 1 (HIF1) stimulates expression of SIAH1/2, two E3 ubiquitin ligases. SIAH1/2 then promote ubiquitination and degradation of LATS2, leading to YAP/TAZ activation (Ma et al., 2015; Xiang et al., 2014) (Figure 3). In addition, HIF1 directly induces transcription of TAZ (Xiang et al., 2014), and YAP interacts with and stabilizes HIF1 to enhance transcription of HIF1 target genes (Ma et al., 2015).

YAP/TAZ are potent stimulators of cell growth and proliferation, which are energy-consuming processes. The regulation of Hippo signaling by AMPK suggests that metabolic status can function as a checkpoint for growth-promoting activity of YAP/TAZ. Under conditions of nutrient deprivation or energy crisis, YAP/TAZ activity needs to be restricted to prevent energy exhaustion caused by anabolic processes. Oxygen also plays a critical role in cellular metabolism, and hypoxia is involved in different pathological processes such as cancer. The link between hypoxia and YAP/TAZ activity indicates a role of YAP/TAZ in mediating oncogenic effect of hypoxia.

Actin Cytoskeleton Integrates Upstream Signals

The actin cytoskeleton and Rho GTPases are not only important in maintaining cell morphology, but also play important roles in regulating cell proliferation and differentiation (Jaffe and Hall, 2005). Manipulation of the actin cytoskeleton, such as overexpression of Rho GTPases or inhibition of Rho by C3 toxin, dramatically modulates YAP/TAZ activity (Dupont et al., 2011; Yu et al., 2012; Zhao et al., 2012). Rho GTPases and changes in actin cytoskeleton dynamics have been demonstrated to be key mediators of mechanical cues, GPCR ligands, and cell attachment in regulating the Hippo pathway (Figure 3). Consistently, deletion of different regulators of the actin cytoskeleton impacts YAP and Yki activity. For instance, loss of actin-capping proteins or the *Capulet* gene (which inhibits actin polymerization) in *Drosophila* results in Yki activation and tissue overgrowth (Fernández et al., 2011; Sansores-Garcia et al., 2011). Similarly, knockdown of actin-capping proteins or filamentous actin (F-actin)-severing proteins (cofilin or gelsolin) in mammalian cells also leads to YAP activation (Aragona et al., 2013). In general, Rho GTPase activity and F-actin appear to activate YAP/TAZ, whereas destabilization of F-actin inhibits YAP/TAZ (Dupont et al., 2011; Miller et al., 2012; Wada et al., 2011; Yu et al., 2012; Zhao et al., 2012).

Spectrin proteins, in association with short actin filaments, are organized into an elastic polygonal meshwork that lines the intra-

cellular side of the plasma membrane. In epithelial cells, localization of the spectrin network is usually polarized and present in both apical and basolateral domains. In addition to a supporting role for cell structure, the spectrin network may transmit diverse signals from cell microenvironment to regulate cellular functions (Bennett and Gilligan, 1993). Recently, three independent studies revealed a regulatory role of the spectrin network on the Hippo pathway, as disruption of the spectrin network in multiple *Drosophila* tissues leads to activation of Yki and tissue outgrowth (Deng et al., 2015; Fletcher et al., 2015; Wong et al., 2015). A similar phenomenon was also observed in mammalian cells to support a role of spectrin in YAP regulation (Figures 2 and 3). Therefore, spectrins or associated actin filaments may function as a major node for integrating upstream signals, such as mechanical cues.

Despite its apparent importance, it remains unclear how the actin cytoskeleton regulates activity of Hippo pathway core kinases. One possibility is that multiple Hippo pathway components are enriched at the apical domain via an actin-mediated mechanism that facilitates signal transduction, and actin remodeling may reinforce or disrupt the clustering of Hippo pathway components. In another scenario, the Hippo pathway could be regulated by contractile actomyosin and cellular tension (Figure 3). Inhibition of tension-related enzymes, such as non-muscle myosin (by Blebbistatin), Rho kinases (ROCK, by Y27632), and myosin light-chain kinase (by ML-7), results in YAP/TAZ inhibition (Dupont et al., 2011; Wada et al., 2011). However, how tension is sensed by the Hippo pathway kinases is not fully understood. Treating cells with small molecules to inhibit cellular tension may also affect actin dynamics, so it is difficult to separate the effects of actin remodeling and tension generated by actomyosin. However, it has been proposed that, in *Drosophila*, Ajuba (Jub) plays a critical role in tension-induced Yki regulation. In response to high tension, Jub recruits Wts to intercellular junctions via interactions with α -catenin, thereby inhibiting Wts activity (Rauskolb et al., 2014). In addition, JNK activation upon mechanical strain has also been shown to repress LATS1 (Codelia et al., 2014).

Although it has been initially reported that YAP regulation by mechanical signals is independent of LATS1/2, recent studies suggest that LATS1/2 are involved in YAP/TAZ regulation by the actin cytoskeleton (Miller et al., 2012; Wada et al., 2011; Yu et al., 2012; Zhao et al., 2007, 2012). This discrepancy may be due to incomplete LATS1/2 depletion in knockdown experiments reported in earlier studies. Supporting a role for LATS1/2, the phosphorylation status and in vitro kinase activity of LATS1/2 are clearly regulated upon actin cytoskeleton rearrangement, and the kinetics are similar to that of YAP/TAZ phosphorylation (Yu et al., 2012; Zhao et al., 2012). The regulation of Yki/YAP/TAZ by spectrins or cell attachment/detachment has been shown to be dependent on Wts and LATS1/2 (Deng et al., 2015; Fletcher et al., 2015; Zhao et al., 2012). Moreover, during the preimplantation stage of mouse embryo development, LATS1/2 are essential for YAP regulation by small molecules targeting the actin cytoskeleton (Kono et al., 2014). Collectively, mechanical signals most likely act through Wts and LATS1/2 to influence the activity and function of Yki and YAP/TAZ.

Negative Feedback Regulation and Crosstalk

High Yki or YAP/TAZ activity, especially over the long term, results in tissue overgrowth or cancer (see below). Thus, the physiological fluctuation of Hippo signaling must be tightly controlled to avoid detrimental effects. In *Drosophila*, regulation of the Hippo pathway is fine-tuned by a built-in negative-feedback loop in which activation of Yki turns on the expression of upstream regulators, including Four-jointed, Ex, Mer, Kibra, and Wts (Cho et al., 2006; Genevet et al., 2010; Hamaratoglu et al., 2006; Jukam et al., 2013). Consistently, a similar negative feedback mechanism also exists in mammalian cells. YAP/TAZ directly induce the transcription of *NF2*, *LATS2*, and probably *MST1*, leading to *LATS1/2* activation and YAP/TAZ inhibition (Chen et al., 2015b; Dai et al., 2015; Moroishi et al., 2015b). In addition, YAP/TAZ induce expression of angiomotin-like protein 2 (AMOTL2), a negative regulator of YAP (Mohseni et al., 2014; Zhao et al., 2008). This negative feedback loop is critical for maintaining the proper transient activation of YAP/TAZ upon stimulation. By these mechanisms, YAP and TAZ antagonize each other and may provide a buffer for fluctuations in Hippo pathway activity to ensure tissue homeostasis. When this negative feedback is disrupted, dysregulation of the Hippo pathway may lead to tumorigenesis.

Elucidating the Hippo pathway regulation and function is complicated by crosstalk between the Hippo pathway and many other developmental signaling pathways (reviewed in Varelas, 2014). Besides Wnt, YAP/TAZ may also be regulated by sonic hedgehog (SHH) signaling (Fernandez-L et al., 2009). On the other hand, YAP/TAZ has been shown to regulate the expression of ligands for Wnt, Shh, transforming growth factor β (TGF- β), JAK-STAT, EGFR, and Notch pathways (reviewed in Yu et al., 2015). The extensive signaling crosstalk may create a microenvironment rich in different factors, which in turn regulates cell fate through autocrine or paracrine mechanisms in both cell-autonomous and cell-non-autonomous manners.

Hippo Pathway in Early Embryonic Development

YAP/TAZ are critical during early embryonic development. Although TAZ knockout mice are viable, YAP knockout mice die at E8.5, and blastomeres stop dividing before the morula (16–32 cells) stage when YAP and TAZ are both deleted (Hossain et al., 2007; Makita et al., 2008; Morin-Kensicki et al., 2006; Nishioka et al., 2009; Tian et al., 2007). Therefore, the role of YAP/TAZ in early development is partially overlapping.

The first cell fate specification during embryogenesis occurs during preimplantation stage, in which the trophectoderm (TE) and inner cell mass (ICM) are formed. The TE consists of the outer cells of the blastocyst and forms extraembryonic tissues, while the ICM contains the inner cells of the blastocyst and give rise to the embryo proper and other tissues. The formation of the TE and ICM is mainly due to the position or polarity of cells in the morula, in which the inner and apolar cells form the ICM, while the outer and polar cells give rise to the TE (Sasaki, 2015). As early as the 16-cell stage, YAP/TAZ already show differential subcellular localization between inner and outer cells, and this difference lasts to the blastocyst stage. The different distribution of YAP/TAZ in the TE and ICM results in different gene expression signatures, especially the induction of TE-spe-

cific genes, such as *Cdx2* in outer cells, thus directing cell fate specification (Nishioka et al., 2009). Indeed, mouse embryos with TEAD4 knockout failed to develop TE cells, with all cells differentiating into ICM. On the other hand, depleting *LATS1/2*, *NF2*, or *AMOT/AMOTL2* turns all cells into TE lineage, and these embryos failed to develop ICM-derived tissues (Cockburn et al., 2013; Hirate et al., 2013; Lorthongpanich et al., 2013). These results suggest that the Hippo pathway plays a key role in early embryonic cell specification.

Hippo Signaling in Organ Size Control and Tissue Homeostasis

The effect on organ size is the best-known physiological function of the Hippo pathway. In *Drosophila*, mutation of Hippo pathway kinases (*hpo* and *wts*) or upstream regulators (*ex*, *mer*, *kibra*, *ft*, etc.) leads to overgrowth of organs such as eyes, wings, or other appendages, and transgenic expression of *yki* results in a similar phenotype (reviewed in Halder and Johnson, 2011; Pan, 2010). The increased tissue/organ size is mainly due to Yki-induced cell proliferation and survival.

The effect of the Hippo pathway on organ size is highly conserved in mammals, as revealed by many studies performed in mice (summarized in Table 2). For instance, liver-specific transgenic *Yap* expression in mice can produce a dramatically enlarged liver. Remarkably, the liver returns to its normal size via apoptosis once *Yap* overexpression is turned off (Camargo et al., 2007; Dong et al., 2007). Similarly, liver-specific knockout of *Mst1/2*, *Sav1*, or *Nf2* also results in liver enlargement (Yin et al., 2013; Zhang et al., 2010; Zhou et al., 2009). The mouse embryonic heart is enlarged when *Sav1*, *Mst1/2*, or *Lats1/2* is deleted, and proliferation or apoptosis of cardiomyocytes is sensitive to genetic manipulation of *Yap* (Del Re et al., 2013; Heallen et al., 2011; Lin et al., 2014; von Gise et al., 2012; Xin et al., 2013; Xin et al., 2011).

However, not all organs are equally sensitive to Hippo pathway mutations. For instance, *Mst1/2* knockout results in dramatic overgrowth of liver, heart, stomach, and spleen, but not of kidney, lung, or limbs (Song et al., 2010). A possible explanation is that there are *MST1/2*-independent regulators of YAP/TAZ in these tissues. In the breast and intestine, tissue-specific deletion of *Yap* does not result in any defects in tissue structure or size (Cai et al., 2010; Chen et al., 2014). *Yap* knockout in the mouse liver leads to bile duct defects, but not a general reduction of liver size. This lack of effect could be due to the presence of TAZ, which should be activated due to the loss of feedback inhibition upon *Yap* deletion. Alternatively, the Hippo pathway, or YAP/TAZ activity, may be negligible for the size control of some organs. Nevertheless, the organ-specific effects of the Hippo pathway in size control may suggest that different principles are utilized for size regulation in different organs. For example, organ size could be determined by proliferation of differentiated cells or the number of progenitor cells in a pre-allocated pool.

Physiological signals upstream of the Hippo pathway important in organ size determination have yet to be identified. Mechanical force or tension may change as a result of organ growth and may inhibit YAP/TAZ when the organ reaches its final size. Alternatively, organ size may be restricted/induced by a soluble factor via autocrine/paracrine mechanisms, and the

Table 2. Physiological and Pathological Functions of Hippo Pathway Genes in Mice

Organ	Phenotypes	References
Liver	<i>Yap</i> -inducible expression results in hepatomegaly in a reversible manner and, in long term, leads to development of liver tumors.	Camargo et al., 2007; Dong et al., 2007
	<i>Nf2</i> , <i>Sav1</i> , or <i>Mst1/2</i> deletion causes hepatomegaly and results in hepatocellular carcinoma, cholangiocarcinoma, or bile duct hamartoma. <i>Mob1a/b</i> deletion also causes liver cancer.	Benhamouche et al., 2010; Lee et al., 2010; Lu et al., 2010; McClatchey et al., 1998; Nishio et al., 2012; Song et al., 2010; Yin et al., 2013; Zhang et al., 2010; Zhou et al., 2009
	<i>Yap</i> deletion causes bile duct defect, and YAP activity is involved in liver regeneration upon tissue damage.	Bai et al., 2012; Su et al., 2015; Yimlamai et al., 2014; Zhang et al., 2010
	<i>Lkb1</i> -deficiency-induced liver overgrowth is dependent on YAP activation.	Mohseni et al., 2014
Intestine	<i>Yap</i> transgenic expression causes intestinal dysplasia.	Camargo et al., 2007
	Deletion of <i>Sav1</i> or <i>Mst1/2</i> in mouse intestine results in expansion of progenitor cells, colonic polyps, or adenoma.	Cai et al., 2010; Lee et al., 2008; Zhou et al., 2011
	Deletion of <i>Yap</i> in mouse intestine shows no obvious phenotype but affects regeneration upon tissue damage.	Cai et al., 2010
	<i>Yap</i> transgenic expression (intestine specific) results in rapid loss of intestinal crypts by repressing Wnt signaling.	Barry et al., 2013
	<i>Apc</i> -deletion-induced expansion of intestinal crypts in a YAP/TAZ-dependent manner.	Azzolin et al., 2014; Cai et al., 2015
Skin	Deletion of <i>Sav1</i> or transgenic expression of <i>Yap</i> leads to expansion of basal progenitor cells and skin thickening; long-term activation of YAP results in squamous cell carcinoma. <i>Mob1a/b</i> deletion also causes skin cancer.	Camargo et al., 2007; Lee et al., 2008; Nishio et al., 2012
	Deletion of <i>Mst1/2</i> shows no clear phenotype. Deletion of <i>Ctnn1</i> (α -Catenin) leads to keratinocyte hyperproliferation and squamous cell carcinoma likely mediated by YAP activation.	Schlegelmilch et al., 2011
	<i>Gnas</i> (Gs) deletion causes basal-cell carcinoma partially dependent on YAP activation.	Iglesias-Bartolome et al., 2015
Heart	Deletion of <i>Sav1</i> , <i>Mst1/2</i> , or <i>Lats2</i> at embryonic stage or transgenic expression of active <i>Yap</i> mutant results in hyperproliferation of cardiomyocytes and enlargement of heart.	Del Re et al., 2013; Heallen et al., 2011; Lin et al., 2014; von Gise et al., 2012; Xin et al., 2013; Xin et al., 2011
	Deletion of <i>Yap</i> leads to heart hypoplasia, and more severe phenotype is observed when both <i>Yap</i> and <i>Taz</i> are deleted.	von Gise et al., 2012; Xin et al., 2011
	In adult heart, high YAP/TAZ activity enhances heart regeneration following cardiac damages such as myocardial infarction.	Heallen et al., 2013; Lin et al., 2014; Xin et al., 2013
Kidney	<i>Taz</i> deletion causes polycystic kidney disease, whereas <i>Yap</i> deletion leads to hypoplastic kidneys with severe defect in nephron morphogenesis.	Hossain et al., 2007; Makita et al., 2008; Reginensi et al., 2013; Tian et al., 2007
	Kidney with <i>Mst1/2</i> or <i>Sav1</i> deletion appears normal.	Reginensi et al., 2013; Song et al., 2010
Lung	<i>Taz</i> -deleted mice display abnormal alveolar structures.	Makita et al., 2008; Mitani et al., 2009; Tian et al., 2007
	<i>Mst1/2</i> deletion leads to disrupted lung structures and neonatal lethality, which is dependent on high YAP/TAZ activity. <i>Mst1/2</i> deletion in adult lung bronchiolar epithelial cells results in airway hyperplasia and altered differentiation. YAP appears critical in regulating proximal-distal patterning of the lung, and a decrease in YAP activity ensures epithelial cells differentiation.	Lange et al., 2015; Lin et al., 2015a; Mahoney et al., 2014
	<i>Mob1a/b</i> deletion causes lung tumor.	Nishio et al., 2012
	Forkhead box A2 (FOXA2) has also been shown critical in mediating the effect of <i>Mst1/2</i> in lung development.	Chung et al., 2013

(Continued on next page)

Table 2. Continued

Organ	Phenotypes	References
Pancreas	The effect of Hippo pathway is mainly in the exocrine compartment of the pancreas. During postnatal stage, deletion of <i>Mst1/2</i> increases the ratio of ductal and acinar cells and leads to pancreatitis-like autodigestion and a reduced size of pancreas.	Gao et al., 2013 ; George et al., 2012
	<i>Kras</i> (K12D)-mutant-induced pancreatic ductal adenocarcinoma requires YAP activity.	Zhang et al., 2014a
Nervous System	<i>Nf2</i> deletion causes an expansion of the neural progenitor pool and results in enlargement of the cortical hem, malformation of hippocampus (at late embryogenesis), and thickening of the neurocortex. <i>Nf2</i> deletion also affects development of corpus callosum, in which Yap-mediated overexpression of SLIT2 disrupts callosal axon pathfinding.	Lavado et al., 2013 ; Lavado et al., 2014
Mammary glands	<i>Yap</i> and <i>Sav1</i> are dispensable in mammary glands development. During pregnancy, <i>Yap</i> deletion results in hypoplasia and reduced alveolar structures; on the other hand, <i>Sav1</i> deletion or transgenic expression of <i>Yap</i> prevents terminal differentiation of mammary cells.	Chen et al., 2014
	<i>Mob1a/b</i> deletion causes breast tumor.	Nishio et al., 2012
	<i>Yap</i> deletion delays mammary tumor growth induced by polyoma middle T antigen (PyMT).	Chen et al., 2014
Muscle	<i>Yap</i> overexpression promotes proliferation of satellite cells and represses their differentiation. YAP downregulation reduces basal skeletal muscle fiber size, and YAP activity is required to relieve neurogenic muscle atrophy following injuries.	Judson et al., 2012 ; Watt et al., 2015

concentration of this factor is controlled by organ size. These two models are not mutually exclusive, and further investigations are needed to address how the Hippo pathway senses physiological cues to modulate organ size.

High YAP/TAZ activity has been observed in the stem or progenitor cells of multiple tissues, suggesting a role for YAP/TAZ in stem cell maintenance. For example, YAP is highly nuclear in basal progenitor cells and in intestinal stem cells localized at the crypt base ([Barry et al., 2013](#); [Camargo et al., 2007](#); [Schlegel-milch et al., 2011](#); [Zhang et al., 2011](#)). Activation of YAP, either by transgenic expression of YAP or deletion of upstream regulators, usually results in expansion of progenitor cells, impaired cell differentiation, and hyperplasia of target tissues such as intestine, liver, skin, and nervous system ([Cai et al., 2010](#); [Camargo et al., 2007](#); [Cao et al., 2008](#); [Lee et al., 2008, 2010](#); [Lu et al., 2010](#); [Zhou et al., 2011](#)).

The role of YAP/TAZ on cell proliferation and stem cell expansion suggests a critical function of YAP in normal tissue development and homeostasis. Indeed, tissue-specific deletion of *Yap* results in abnormalities of the heart, skin, and kidney ([Reginensi et al., 2013](#); [Schlegel-milch et al., 2011](#); [von Gise et al., 2012](#); [Xin et al., 2011](#)). However, mammary glands and the intestine remain relatively normal upon *Yap* deletion ([Cai et al., 2010](#); [Chen et al., 2014](#); [Zhou et al., 2011](#)). These findings suggest that YAP is required for development and homeostasis of some, but not all, tissues in mice. In human, *TEAD1* mutations are found in Sveinsson chorioretinal atrophy, a disease characterized by chorioretinal degeneration, and Aicardi syndrome, a congenital neurodevelopmental disorder ([Fossdal et al., 2004](#); [Schrauwen et al., 2015](#)). In addition, loss-of-function mutations of YAP

have been identified in both isolated and syndromic optic fissure closure defects ([Williamson et al., 2014](#)). Hence, the loss of TEAD-mediated YAP transcriptional activity plays a role in some degeneration-related disorders in humans.

Even though it is not required for development and normal homeostasis of some tissues, YAP activity is critical for tissue regeneration upon certain types of damage. For example, *Yap* deletion severely compromises pregnancy-induced mammary tissue growth, although virgin mammary development was normal ([Chen et al., 2014](#)). Likewise, in wild-type mice, the intestines can effectively regenerate following colitis induced by dextran sulfate sodium (DSS) treatment; however, the regenerative capability is severely impeded in conditional *Yap* knockout mice ([Cai et al., 2010](#)). Similar results have also been observed in *Drosophila* midgut regeneration upon DSS-induced injury ([Karpowicz et al., 2010](#); [Ren et al., 2010](#); [Shaw et al., 2010](#)). Normally the liver regenerates efficiently following liver damage. For instance, after partial hepatectomy, hepatocytes start to proliferate to restore liver mass in a few days, and in this process, YAP activity is induced and is most likely required for complete liver regeneration ([Grijalva et al., 2014](#); [Su et al., 2015](#); [Wu et al., 2013](#); [Yimlamai et al., 2014](#)). In contrast to the intestine and liver, tissue regeneration of the adult heart is very limited. However, inactivation of the Hippo pathway or transgenic expression of *Yap* restores some myocardial regenerative capability, although the efficiency is low. In contrast, cardiac-specific deletion of *Yap* impedes regeneration of the neonatal heart ([Heallen et al., 2013](#); [Lin et al., 2014](#); [Xin et al., 2013](#)). Taken together, these results indicate that YAP plays a significant role in regeneration of multiple tissues.

Hippo Signaling in Cancer

Long-term YAP activation, such as transgenic expression of *Yap* in the mouse liver, results in cell transformation and tumor development (Dong et al., 2007), indicating the power of the Hippo pathway in cancer initiation and progression. Evidence of the Hippo pathway in tumorigenesis based on mouse models is summarized in Table 2, which generally supports an oncogenic role for YAP/TAZ, as well as a tumor-suppressive function for Hippo pathway upstream components.

The tumor-promoting activity of YAP is largely dependent on a TEAD-mediated transcription program, as YAP-induced liver cancer is fully blocked by expression of a dominant-negative TEAD that is able to sequester both YAP and TAZ (Liu-Chittenden et al., 2012). At the cellular level, YAP activation is important for cell proliferation, survival, migration, and invasion. High YAP or TAZ activity enables the cell to escape contact inhibition and anoikis and to support anchorage-independent growth (Chan et al., 2008; Zhao et al., 2007, 2012). YAP induces expression of ZEB1/2 to stimulate the epithelial-to-mesenchymal transition (EMT), which is a key step for tumor metastasis (Gao et al., 2014; Lei et al., 2008; Liu et al., 2010b; Overholtzer et al., 2006). In addition, YAP is able to promote genomic instability (Fernandez-L et al., 2012), and TAZ is required to sustain self-renewal and the tumor-initiation capacity of breast cancer stem cells (Cordenonsi et al., 2011).

Accumulating evidence suggests that the Hippo pathway is dysregulated in many human cancers. Elevated YAP/TAZ expression or nuclear enrichment of YAP/TAZ has been observed in many types of cancers, including liver, breast, lung, colon, ovary, and others (Chan et al., 2008; Moroishi et al., 2015a; Steinhardt et al., 2008). However, the majority of cancers with high YAP/TAZ activity have not been linked to genetic mutations of the Hippo pathway, and the overall genetic alteration rate of Hippo pathway components in human cancer is relatively low (Table 3).

One well-characterized example of a Hippo pathway mutation associated with cancer is in *NF2*, which causes neurofibromatosis 2 lesions, including schwannomas and meningiomas (Xiao et al., 2003). Moreover, inactivating *NF2* mutations are also observed in 40%–50% of malignant mesothelioma (Sekido, 2011). Importantly, even heterozygous deletion of *Yap* completely blocks liver tumorigenesis induced by *Nf2* knockout in mice, indicating that YAP activation is the major mechanism mediating the tumorigenic potential of *Nf2* mutations (Zhang et al., 2010).

YAP gene amplification may contribute to a portion of hepatocellular carcinomas, medulloblastomas, and esophageal squamous cell carcinomas (Fernandez-L et al., 2009; Overholtzer et al., 2006; Song et al., 2014; Zender et al., 2006). Gene fusions involving YAP or TAZ have also been discovered in human cancers. Remarkably, virtually all epithelioid hemangioendotheliomas contain gene fusions of *TAZ-CAMTA1*, *TAZ-FOSB*, or *YAP-TFE3* (Antonescu et al., 2013; Errani et al., 2011; Flucke et al., 2014; Tanas et al., 2011). In addition, YAP gene fusions with *MAML1* or *C11orf95* have been discovered in a subset of ependymal tumors (Pajtl et al., 2015; Parker et al., 2014). It is worth noting that, in both epithelioid hemangioendotheliomas and ependymal tumors, all YAP/TAZ fusion proteins retain their N-terminal TEAD-binding domain but lose the C-terminal trans-

activation domain. These observations suggest that these YAP fusions may still bind to and activate the TEAD-dependent transcriptional program to promote tumorigenesis. Indeed, neural stem cells carrying the *YAP-C11orf95* fusion gene can effectively form brain tumors when grafted into mice (Parker et al., 2014). In addition, a familial YAP point mutation (R331W) has also been reported to correlate with a high incidence of lung adenocarcinomas (Chen et al., 2015a).

Aberrant GPCR signaling often results in tumorigenesis, so it is possible that GPCR dysregulation can cause cancer by activating YAP/TAZ. *GNAQ*- or *GNA11*- (encoding $G\alpha_q$ or $G\alpha_{11}$, respectively) activating mutations have been identified in ~80% of uveal melanomas and function as driver mutations (Van Raamsdonk et al., 2009; Van Raamsdonk et al., 2010). Recent studies showed that YAP is constitutively activated in *GNAQ*- or *GNA11*-mutated uveal melanomas, and the high YAP activity contributes to tumor growth (Feng et al., 2014; Yu et al., 2014). In mice, deletion of *Gnas* (encoding $G\alpha_s$) in skin stem cells initiates basal-cell carcinogenesis, which is partially dependent on YAP (Iglesias-Bartolome et al., 2015). Moreover, expression of a viral GPCR induces tumorigenesis in Kaposi's sarcoma, where YAP/TAZ also play a critical role (Liu et al., 2015).

LATS1/2 mutations or gene fusion have been sporadically identified in different cancers, which may lead to YAP/TAZ activation (Table 3). In addition, crosstalk with other cancer-related signaling pathways also likely contributes to high YAP/TAZ activity in cancers that have no mutations of Hippo pathway components. For example, *KRAS*, *APC*, and *LKB1* mutations have all been reported to activate YAP/TAZ (Azzolin et al., 2014; Gao et al., 2014; Mohseni et al., 2014; Zhang et al., 2014a).

YAP/TAZ activity is also linked to drug resistance and cancer relapse. Cultured breast cancer cells with high YAP/TAZ activity show resistance to drugs such as taxol, 5-fluorouracil, and doxorubicin (Cordenonsi et al., 2011; Lai et al., 2011; Touil et al., 2014). Furthermore, lung and colon cancer cells with high YAP activity are resistant to RAF- and MEK-targeted therapies (Lin et al., 2015b). Tamoxifen is commonly used to treat estrogen-receptor- (ER, a nuclear receptor) positive breast cancer; however, some ER-positive breast cancers are insensitive to tamoxifen. Recently, tamoxifen has been shown to activate YAP/TAZ by stimulating the membrane estrogen receptor GPER, a GPCR. Therefore, activation of GPER by tamoxifen or estrogen may contribute to tumor growth and drug resistance (Zhou et al., 2015). Amplification of the YAP gene has been associated with cancer relapse in *KRAS*-driven colon and pancreatic cancers (Kapoor et al., 2014; Shao et al., 2014). Thus, inhibition of YAP/TAZ will not only target tumor initiation and progression, but also potentially sensitize cancer cells to chemotherapies and prevent cancer relapse.

Notably, in contrast to its oncogenic function in most solid tumors, YAP seems to play a tumor-suppressor role in hematological cancers. The YAP gene locus is frequently deleted in hematological cancer, and expression of YAP or inhibition of MST1 leads to growth inhibition and increased apoptosis (Cottini et al., 2014). Currently, the underlying mechanism responsible for this tumor-suppressor function of YAP in hematological cancers is not well understood.

Table 3. Genetic Alterations of Hippo Pathway Genes in Human Cancers

Gene	Alteration	Cancer type	References
<i>NF2</i>	mutation or deletion	mesothelioma neurofibromatosis type 2 (schwannoma, meningioma)	Sekido, 2011 Rouleau et al., 1993
<i>LATS1/2</i>	gene fusion (<i>LATS1-PSEN1</i>)	mesothelioma	Miyanaaga et al., 2015
	<i>LATS2</i> deletion	mesothelioma	Murakami et al., 2011
	<i>LATS1/2</i> mutations	sporadic in different cancers	Yu et al., 2013b
<i>YAP</i>	amplification	hepatocellular carcinoma medulloblastoma esophageal squamous cell carcinoma	Fernandez-L et al., 2009 Overholtzer et al., 2006 Song et al., 2014 Zender et al., 2006
	mutation (R331W)	lung adenocarcinoma	Chen et al., 2015a
	gene fusion (<i>YAP-TFE3</i> , <i>YAP-ESR1</i> , <i>YAP-C11orf95</i> , and <i>YAP-MAMLD1</i>)	epithelioid hemangioendothelioma luminal breast cancer ependymal tumors	Antonescu et al., 2013 Flucke et al., 2014 Li et al., 2013 Pajtlar et al., 2015 Parker et al., 2014
	deletion	hematological cancer	Cottini et al., 2014
	gene fusion (<i>TAZ-CAMTA1</i> and <i>TAZ-FOSB</i>)	epithelioid hemangioendothelioma	Errani et al., 2011 Flucke et al., 2014 Tanas et al., 2011
<i>GNAQ/GNA11</i>	activating mutation	uveal melanoma	Van Raamsdonk et al., 2009 Van Raamsdonk et al., 2010

Therapeutic Targeting of Hippo Signaling

The core Hippo pathway is a kinase cascade, and protein kinases are usually druggable. Thus, inhibitors for MAP4K4, MST1/2, or LATS1/2 may be developed to induce YAP/TAZ activity and facilitate the process of wound healing, tissue repair, or regeneration and possibly for treating degenerative diseases (Figure 4). For example, temporal inhibition of MST1/2 or LATS1/2 may promote myocardial regeneration or survival that would be beneficial for heart attack patients. It is also possible that inhibitors of MST1/2 or LATS1/2 could be used for treating hematological cancers.

Generally, MST1/2 and LATS1/2 are tumor suppressors, and inhibition of MST1/2 or LATS1/2 may promote tumor growth in most instances. On the other hand, inhibiting YAP/TAZ activity would offer a new and attractive anti-cancer strategy (Park and Guan, 2013). The function of YAP/TAZ is primarily mediated by TEADs, so small molecules disrupting the YAP/TAZ-TEAD interaction will function as YAP/TAZ inhibitors. Indeed, porphyrin family molecules, especially verteporfin, are able to disrupt the interaction between YAP/TAZ and TEADs, and verteporfin can block transcription of YAP/TAZ target genes and suppress liver overgrowth induced by YAP overexpression or NF2 inactivation in mice (Liu-Chittenden et al., 2012). However, verteporfin has general cellular toxicity and low aqueous solubility. Based on structural information from the YAP-TEAD and VGLL4-TEAD complex, a polypeptide termed “super-TDU” has been designed to block YAP-TEAD interaction and has been shown to suppress tumor growth in mouse models (Jiao et al., 2014).

It is challenging to design direct activators for protein kinases. However, LATS1/2 may be activated indirectly by molecules tar-

geting their upstream regulators. The very first small molecule (dobutamine) identified with an inhibitory effect on YAP is an antagonist of a GPCR receptor (Bao et al., 2011). Since then, many indirect inhibitors for YAP/TAZ have been identified, including phosphodiesterase inhibitors rolipram and ibudilast (Yu et al., 2013a). The Rho family GTPases have a strong inhibitory effect on LATS1/2, and membrane localization is important for Rho cellular function. Indeed, mevalonate metabolic pathway inhibitor statins can block membrane translocation of Rho GTPases and indirectly inhibit YAP/TAZ activity (Mi et al., 2015; Sorrentino et al., 2014; Wang et al., 2014). It will be interesting to test whether these drugs are effective in suppressing tumor growth in mouse models and to perform epidemiologic studies to determine whether patients using statins or rolipram have a lower incidence of cancer. Given the important function of the Hippo pathway in regulating cell proliferation and tissue homeostasis, it represents an exciting and previously unexplored field for cancer therapy.

Outstanding Questions

Despite the rapid research progress in the Hippo pathway, some key questions remain unanswered, and new questions are emerging. Listed below are some of the key questions in the Hippo field.

- (1) What are the molecular mechanisms regulating MST1/2 activity? Many upstream signals have been convincingly shown to regulate LATS1/2 phosphorylation and kinase activity. However, neither MST1/2 phosphorylation nor its kinase activity is strongly modulated by upstream

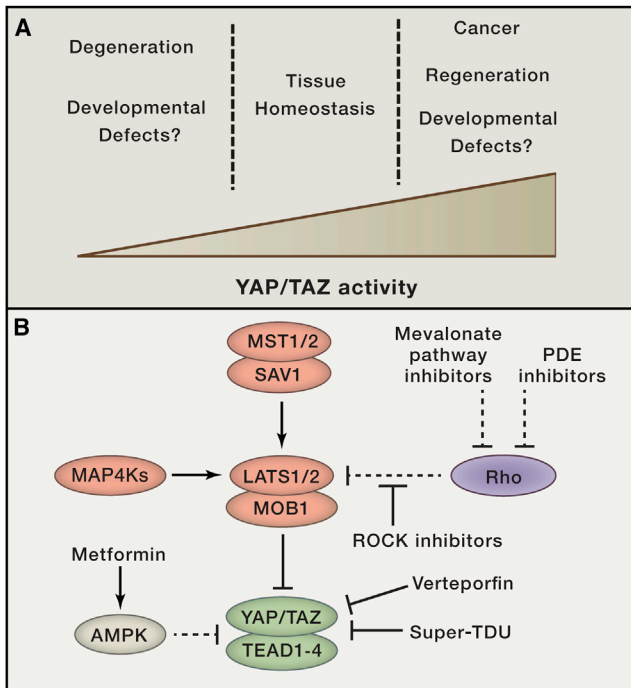


Figure 4. Therapeutic Targeting of the Hippo Pathway

(A) Potential roles of YAP/TAZ activity in tissue development and diseases. A confined window of YAP/TAZ activity is required for normal tissue development and homeostasis.

(B) Strategies for targeting YAP/TAZ activity. Inhibitors for MST1/2, MAP4K4, and LATS1/2 can activate YAP/TAZ. YAP/TAZ-TEAD interaction may be disrupted by small molecules directly (Verteporfin) or AMPK activators (Metformin). Small molecules inhibiting Rho-family GTPases or ROCK can indirectly activate LATS1/2, leading to YAP/TAZ inhibition.

signals. *Drosophila* misshapen (a member of MAP4K in mammals) acts upstream of Wts. An interesting question is whether MST1/2 and MAP4Ks mediate different or similar upstream signals to activate LATS1/2.

- (2) Where are Hippo pathway components localized in mammalian cells? This may be key to understanding how the Hippo pathway is regulated in response to upstream signals. It is obvious that phosphorylated YAP/TAZ are enriched in the cytoplasm and dephosphorylated YAP/TAZ in the nucleus. However, it is less clear where YAP/TAZ phosphorylation and dephosphorylation occur. A related question is what are the mechanisms underlying YAP/TAZ translocation between the nucleus and cytoplasm?
- (3) How are LATS1/2 regulated by actin remodeling and/or cellular tension? This is one of the key questions in understanding the biochemical mechanism of the Hippo kinase cascade regulation. Accumulating evidence suggests that the actin cytoskeleton and cellular tension play a key role in LATS1/2 regulation and appear to act downstream of many, if not most, upstream signals. The actin cytoskeleton and cellular tension are intertwined, thus the question is: which one plays a more direct role in regulating Hippo core components?

- (4) What is the mechanism of organ size sensing? Although many signals are reported to regulate the Hippo pathway in vitro, so far none have been demonstrated to play a key role in organ size control in vivo. Uncovering this magic signal will solve a key question in developmental biology.
- (5) How does YAP become deregulated in cancer? It is clear that YAP/TAZ activation is observed in a broad spectrum of human cancers, although mutations in Hippo pathway genes are rare. This interesting conundrum indicates that the Hippo pathway may be regulated broadly by many other cancer-driving pathways.

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